Shaam Adam

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Professional Experience

GIS Specialist experienced in web GIS development, spatial data analysis, and Python automation (ArcPy, GeoPandas, Pandas). Builds interactive web applications and dashboards to support data-driven decision-making, with recent work extending into applied AI — deploying locally-hosted LLMs for chatbot tools.

Technical Skills

GIS & Spatial: ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, ArcGIS JavaScript SDK

Programming: Python (ArcPy, Pandas, GeoPandas), SQL (PostgreSQL/PostGIS), R

Web Development: JavaScript, HTML, Arcade, Vite/npm

Applied AI: RAG-based LLM applications, Azure AI Foundry, LiteLLM

Work Experience

U-Spatial – University of Minnesota
Dec. 2022 - Present
GIS Specialist
September 2024 – Present

- Contribute to a greenhouse gas emissions reduction initiative (Global Climate Geodesign Competition) through spatial data analysis and custom tool development
- Develop and publish geoprocessing tools using Python, ArcPy, and the ArcGIS API for Python, deployed via ArcGIS Enterprise Server
- Developed custom web GIS applications using the ArcGIS JavaScript SDK, including bidirectional map-legend interactions and dynamic feature filtering
- Developing a RAG-based AI chatbot to support internal UMN collaboration between researchers, staff, and faculty. Planned migration to Azure AI Foundry and LiteLLM

GIS Technician
June 2023 – September 2024

- Supported the Statewide Lead Service Line Project by cleaning and geocoding large datasets and modifying Python/ArcPy scripts to process spatial and tabular data for analysis. Also supported early development of the Global Climate Geodesign Competition project.

Undergraduate Research Assistant
Dec. 2022 – June 2023

- Developed web layers and HTML content for SDG Hub Site using ArcGIS Online

Education

Master of Geographic Information Science, Minor in Systems and Industrial Engineering
University of Minnesota, College of Liberal Arts
Minneapolis, MN | Expected May 2027

Bachelor of Science in Geography, Minor in Geographic Information Science
University of Minnesota, College of Liberal Arts
Minneapolis, MN | May 2023